



Surgical intervention yields superior results over conservative management in calcaneal-side calcaneofibular ligament rupture

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ABSTRACT

Background: To the best of our knowledge, no study has investigated complete calcaneal-side calcaneofibular ligament (CFL) rupture. This study was performed to compare the clinical and radiographic results of conservative and surgical treatments for acute ankle sprains with complete calcaneal-side CFL rupture.

Methods: This study included 36 patients diagnosed with acute ankle sprain and complete calcaneal-side CFL rupture. The first 14 patients were treated conservatively (group C), and the remaining 22 underwent surgery (group S). Clinical results between the two groups were compared regarding their scores on the Cumberland Ankle Instability Tool (CAIT) and patient satisfaction with the treatment. Radiographic results were evaluated for the rupture status of the CFL on preoperative magnetic resonance imaging and stability of the ankle joint using a stress radiographs with Telos and manual anterior drawer test.

Results: At the last follow-up, the mean CAIT score was significantly higher in group S than in group C ($P < 0.001$). The treatment dissatisfaction rate was significantly higher in group C ($P = 0.003$). In all cases, the proximal migration distance of the ruptured CFL significantly correlated with the incidence of Stener's-like lesions, in which the distal stump of the CFL was displaced over the peroneal tendons ($R = 0.721$, $P < 0.001$). Stress radiographs showed no significant difference in postoperative instability between groups. In the manual anterior drawer test at the last follow-up, the proportion of stable ankles was significantly higher in group S than in group C ($P = 0.02$).

Conclusion: Surgical treatment could improve clinical results and satisfaction in patients with acute ankle sprains and complete calcaneal-side CFL rupture.

Level of evidence: III, Retrospective cohort study

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1. Introduction

The anterior talofibular ligament (ATFL) and calcaneofibular ligament (CFL) serve as the primary stabilizers of the lateral ankle, crucial for resisting varus stress [1]. The ATFL, an extension of the joint capsule, typically consists of superior and inferior fascicles, while the CFL is a cord-like ligament [2,3]. The ATFL is frequently involved in ankle sprains, regardless of the injury grade; in contrast, the CFL is predominantly injured in higher-grade sprains, including grades II and III [4,5]. In higher-grade sprains with CFL injury,

significant biomechanical changes occur, including decreased rotation stiffness, reduced peak torque, substantial alterations in ankle joint contact mechanics, increased talus and calcaneus inversion, and heightened calcaneus medial displacement [6–10].

Although established treatment methods and prognoses exist for ATFL injuries [11–13], there is an ongoing debate surrounding the optimal approach to CFL injuries [7,14,15]. Given their different anatomical implications, it is essential to distinguish between isolated ATFL and combined injuries involving the ATFL and CFL when analyzing clinical outcomes after lateral ligament injury treatment.

Magnetic resonance imaging (MRI) of acute ankle ligament ruptures show that injuries can occur in various forms [16,17]. Recently, it has been revealed that the injury patterns of the CFL are heterogeneous. Kim et al. highlighted that CFL injuries do not adhere to a uniform pattern but can fluctuate considerably depending on

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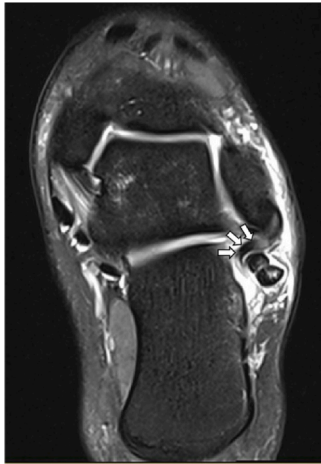


Fig. 1. The calcaneofibular ligament was completely torn at the calcaneal attachment, with proximal migration of the ruptured ligament (arrows) on magnetic resonance imaging.

variables such as the site of the rupture and the degree of ligament displacement [18]. The investigation accentuates the significance of understanding these variations, as they possess direct ramifications for therapeutic methodologies and the prognosis of ankle stability subsequent to injury. In particular, the sleeve avulsion type of CFL injury from calcaneus is documented as a distinctive pattern that had not been thoroughly chronicled previously.

Recent cadaveric studies have revealed that the inferior fascicle of the ATFL is linked to the CFL through arciform fibers, forming the lateral fibulotalocalcaneal ligament complex [5,19–21]. Consequently, significant distal migration of the ruptured ligament complex is less likely when the CFL is completely ruptured at its fibular attachment. Unlike CFL ruptures at the fibular attachment, complete CFL rupture at the calcaneal attachment is rare but can lead to substantial proximal migration, which is challenging to repair (Fig. 1) [19,22]. When proximal migration is severely advanced, the distal stump of the torn CFL can be displaced superficially over the peroneal tendons, similar to Stener's-like lesion in a skier's thumb [23]. Thus, anatomical restoration of the ruptured CFL stump through conservative treatment becomes challenging in complete CFL rupture with severe proximal migration at the calcaneal attachment, increasing the risk of developing chronic ankle instability (CAI) [7,18].

In the present study, CFL rupture at the calcaneal attachment is referred to as calcaneal-side CFL rupture. Currently, limited information is available regarding the treatment options and prognosis for calcaneal-side CFL ruptures, with no prior studies exploring the outcomes of conservative versus surgical management for this specific injury type.

We have experienced unsatisfactory outcomes after conservative treatment for calcaneal-side CFL rupture in athletes in previous studies [23]. Therefore, in this study, we aimed to comprehensively compare the clinical and radiographic outcomes between conservative and surgical treatments for complete calcaneal-side CFL ruptures. We hypothesized that surgical treatment would yield superior clinical and radiographic outcomes compared with conservative treatment for complete calcaneal-side CFL rupture.

2. Materials and methods

2.1. Participants

This study was approved by our hospital's Institutional Review Board, and the requirement for informed consent was waived due to its retrospective design. The authors confirmed calcaneal-side CFL

Table 1
Inclusion and Exclusion Criteria.

Inclusion criteria
Adult patient (age ≥ 18 years)
Acute ankle sprain (defined as < 2 weeks after injury)
Calcaneal-side CFL total rupture in MRI
Tegner scale > 6
Exclusion criteria
Patients with recurrent ankle sprains
Generalized ligamentous laxity [†]
Concomitant deltoid ligament rupture or syndesmotic injury
Concomitant osteochondral injury
Patients with a followed-up period of less than 24 months

CFL, calcaneofibular ligament; MRI, magnetic resonance image

[†]Generalized ligamentous laxity was defined as a positive result of a minimum of 4 of 5 tests according to the Beighton and Horan scoring system

rupture by MRI in a case with diffuse swelling and echymosis on the lateral side of the ankle and heel with prominent tenderness to the calcaneal attachment of CFL. Forty-six patients diagnosed with acute ankle sprain accompanied by complete ruptures of ATFL and calcaneal-side CFL were consecutively treated by a single surgeon between January 2016 and December 2019. After applying the inclusion and exclusion criteria, 36 patients remained eligible for the study (Table 1).

All patients were followed up for a minimum of 24 months after treatment. Two patients with generalized ligamentous laxity, six with deltoid or syndesmotic ligament injuries, and two with acute osteochondral fractures were excluded. Consequently, 36 patients were included in this study. Among them, the initial 14 patients received conservative treatment (group C), whereas the subsequent 22 patients underwent surgical treatment (group S). Age, gender, injured side, injury mechanism, sports level, and follow-up duration were similar between groups C and S. The demographic characteristics of the two groups are listed in Table 2.

2.2. Clinical evaluation

Clinical outcomes were assessed using the Cumberland Ankle Instability Tool (CAIT) [24] at the last follow-up. The CAIT consists of nine items designed to assess different aspects of CAI. These items measure the difficulty in performing various physical activities owing to the CAI of the affected ankle. The total score of the nine items ranges from 0 (indicating severe instability) to 30 (reflecting normal stability). Typically, if the CAIT score ≤ 24 points, the likelihood ratio is 32%, indicating a high probability of experiencing recurrent sprains owing to persistent CAI.

Table 2
Baseline data of Group C and S.

	Group C (14 cases)	Group S (22 cases)	P-value
Age (years)	26.0 $\pm$ 8.5	26.8 $\pm$ 12.7	0.842
Gender			0.311
Male	10 (71.4%)	12 (54.5%)	
Female	4	10	
Side			0.968
Right	9 (64.3%)	14 (63.6%)	
Left	5	8	
Injury mechanism			0.158
Supination	14 (100%)	17 (77.3%)	
Tackle	0	2	
Landing injury	0	3	
Sports level			0.38
Army	0	1	
Athlete	4	10	
Recreational	10 (71.4%)	11 (50%)	
FU periods (weeks)	142.8 $\pm$ 199.9	88.6 $\pm$ 59.8	0.239

FU, follow-up; BMI, body mass index

The timing of return to sports (RTS) after treatment was compared between groups. The timing of RTS was evaluated regarding when the patients could resume running and when they could participate in sports competitions.

At the last follow-up, patients' satisfaction with the treatments was assessed based on their responses, including "very satisfied," "satisfied," "fair," or "dissatisfied." In cases where a patient reported being "dissatisfied," the reasons for the dissatisfaction were assessed.

To compensate for the absence of a preoperative radiographic stress test, a manual anterior drawer test (ADT) was used. In Group C, the test was performed in the outpatient clinic after aspiration of intra-articular hemarthrosis, while in Group S, it was conducted in the operating room under sciatic nerve block. A skilled ankle specialist administered manual ADT preoperatively and at the last follow-up. Instability was graded as normal (grade 0), mild (grade I, slight anterior translation compared with the contralateral ankle), moderate (grade II, significant anterior translation exceeding that of the contralateral ankle with a firm endpoint), or severe (grade III, significant anterior translation without an endpoint), with grades II and III indicating an unstable ankle [25]. Ankle stability was compared between the groups before and after treatment.

2.3. Imaging evaluation

MRI examinations were conducted using 3.0-T scanners (Achieva 3.0 T, Philips Medical System), consisting of fat-suppressed T2 coronal, sagittal, ATFL axial transverse, and 45° oblique coronal images. All MRIs were conducted in the supine position with the foot in a plantigrade position. Images were reviewed by an ankle specialist and a musculoskeletal radiologist. Any disagreements were resolved through discussion and, if necessary, a third musculoskeletal radiologist was consulted. ATFL ruptures were assessed using MRI; if present, the rupture sites were categorized as a fibular attachment, talar attachment, and midsubstance. The difference in the ATFL rupture sites between groups C and S was compared. To evaluate the proximal migration of the distal end of the ruptured CFL, the proximal migration distance (PMD) was measured. PMD was defined as the distance from the rupture site on the calcaneus to the distal end of the ruptured CFL in a 45° oblique coronal plane (Fig. 2) [26]. When the CFL severely migrated proximally and slipped to the outside of the peroneal tendons, it was defined as a Stener's-like lesion (Fig. 3). PMD and incidence of Stener's-like lesions were compared between

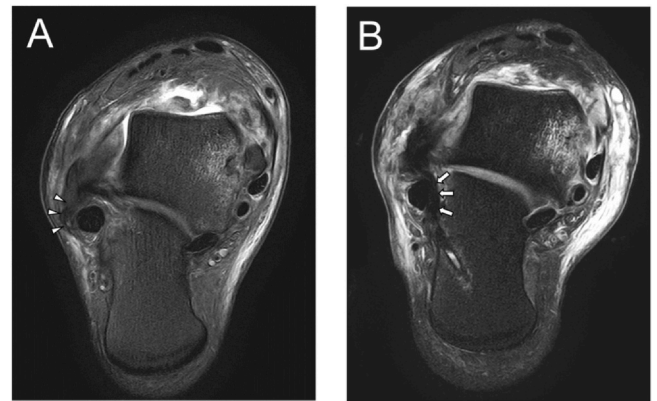


Fig. 3. (A) The ruptured calcaneofibular ligament (CFL) migrated severely proximal and slipped to the outside of the peroneal tendons (arrowheads) on preoperative magnetic resonance imaging (MRI). (B) The torn CFL was repaired with a knotless anchor at the calcaneal attachment site (arrows) on postoperative MRI.

groups C and S. In addition, the relationship between PMD and the incidence of Stener's-like lesions was evaluated in both groups.

All radiographs were obtained at the same facility using the same technique. Talar tilt angle (TTA) and anterior talar displacement (ATD) were measured from stress radiographs using a Telos SE 2000 stress device (ARD Medizin Produkte GmbH, Germany) with 150 N force. In cases of acute ankle sprains, assessing the exact degree of instability using stress radiographs can be challenging because of pain. In addition, in cases of calcaneal-side CFL rupture, stress radiographs, particularly the varus stress test, can potentially exacerbate proximal migration or cause a Stener-like lesion in the torn calcaneal-side CFL stump. Therefore, preoperative stress radiographs were not obtained.

The TTA and ATD were measured using stress radiographs at the last follow-up, and the results were compared between the two groups. TTA was defined as the angle between the articular surfaces of the tibia and talus in the varus stress view. ATD was defined as the shortest distance between the posterior lip of the tibia and the talar dome in the anterior drawer stress view. Each radiographic measurement was performed twice independently by an observer who was not a part of the surgical team, and the average values were calculated for analysis to minimize potential bias.

2.4. Treatment protocol

Conservative treatment consisted of protection, optimal loading, ice, compression, and elevation (POLICE) therapy. A short leg cast was applied at a 90° angle for 2 weeks, and weight bearing was allowed. After 2 weeks, the cast was replaced with a removable ankle brace. The patients were instructed to continue wearing the braces at night. In addition, a combination of physical therapy, manual therapy for edema control, and passive exercise therapy was used to increase the ankle's range of motion. After 4 weeks, the patients resumed exercise to improve muscle strength and body balance. After 8 weeks, the patients were allowed to run within their tolerable pain range.

In all surgical cases, the patients underwent a sciatic nerve block and were placed in the semi-decubitus position. A 4–5 cm incision was made along the anterior surface of the fibula to access and identify the torn ligaments. A torn CFL displaced outside the peroneal tendon was termed a Stener's-like lesion (Fig. 4A) (Video). Typically, the peroneal sheath was dissected, and the peroneal tendons were gently lifted with a retractor to confirm the extent of the torn CFL (Fig. 4B). The torn CFL was then sutured with a non-absorbable suture (FiberWire #2 with a needle; Arthrex, Naples, USA) using the modified Bunnel technique. The suture was secured to the

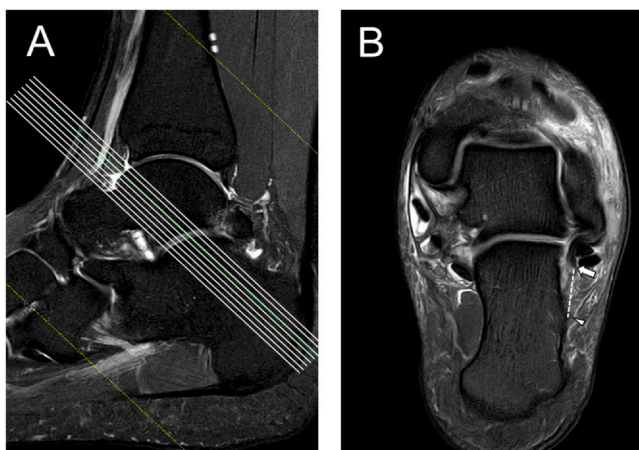


Fig. 2. (A) The proximal migration distance (PMD) was evaluated in a 45° oblique coronal plane of magnetic resonance imaging. (B) The PMD (dotted line) was defined as the distance from the rupture site on the calcaneus (arrowhead) to the distal end of the ruptured calcaneofibular ligament (arrow).

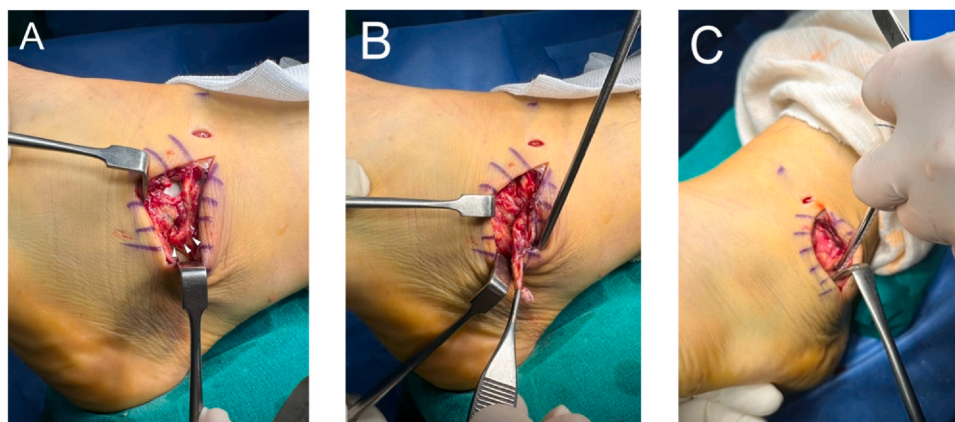


Fig. 4. (A) The Stener's-like lesion in which the calcaneofibular ligament (CFL) (arrowheads) slipped out of its original attachment was observed intraoperatively. (B) The torn CFL was pulled out to reveal the full length of the CFL. (C) The torn CFL was sutured with a non-absorbable suture using the modified Bunnel technique and secured to the calcaneal attachment site with a knotless anchor.

calcaneal attachment site using a knotless anchor (PushLock, Arthrex, Naples, USA) (Fig. 4C). The method for suturing the ATFL varied depending on the location and type of tear. All suture anchors (TruShot, ConMed, USA) were used if a tear occurred at the fibular or talar bony attachment sites. An absorbable suture (Vicryl #2, Ethicon, USA) was used to treat midsubstance tears. The postoperative protocol was the same as that used for group C.

2.5. Statistical analysis

All dependent variables were tested for normality of distribution and equality of variances and were analyzed using nonparametric tests as they had non-normal distributions. The Mann-Whitney U and chi-squared tests were used to compare demographic data between the groups. The Mann-Whitney U test was used to compare the CAIT score and timing of RTS, PMD, TTA, and ATD between the groups. The Fisher's exact test was used to compare the ATFL rupture sites, the occurrence of Stener's-like lesions, and manual ADT between the groups. Regression analysis was used to evaluate the correlation between PMD and the occurrence of Stener's-like lesions. Statistical significance was set at $P < 0.05$. Statistical analyses were performed using SPSS (version 18.0; SPSS Inc., Chicago, IL, USA).

3. Results

The mean CAIT score was significantly better in group S than in group C at the last follow-up ($P < 0.001$). Group C had a high risk of re-sprain, as indicated by a CAIT score of ≤ 24 . The timing of RTS after treatment did not differ between the two groups (Table 3).

Patient satisfaction with the treatment in both groups is shown in Table 3. Seven (50%) patients in group C and one (4.5%) in group S were dissatisfied with the treatment, and the rate of dissatisfaction with the treatment was significantly higher in group C ($P = 0.003$).

Table 3
Clinical results after surgery in Group C and S.

	Group C	Group S	P-value
CAIT score			
Running (weeks)	19.8 $\pm$ 8.4	27.4 $\pm$ 3.2	< 0.001
Timing of RTS			
Running (weeks)	12.2 $\pm$ 8.7	12.4 $\pm$ 14.7	0.976
Sports competition (weeks)	18 $\pm$ 7.5	17.2 $\pm$ 11.8	0.88
Patients satisfaction			0.003
Very satisfied	3	8	
Satisfied	1	10	
Fair	3	3	
Dissatisfied	7 (50%)	1 (4.5%)	

CAIT, Cumberland Ankle Instability Tool; RTS, return to sports

The reasons for dissatisfaction were recurrent ankle sprains in six patients and extreme ankle stiffness in one patient in group C. In group C, three (21.4%) patients underwent surgical treatment for persistent ankle instability and recurrent sprains during follow-up. In group S, one patient was dissatisfied with the treatment because of extreme ankle stiffness postoperatively.

On preoperative MRI, the ATFL rupture site, PMD of the CFL, and incidence of Stener's-like lesions did not differ between groups C and S (Table 4). In all cases, PMD significantly correlated with the incidence of Stener's-like lesions ($R = 0.721$, $P < 0.001$), and these factors also showed significant correlations in groups C ($R = 0.53$, $P = 0.024$) and S ($R = 0.649$, $P < 0.001$).

In stress radiographs, the TTA and ATD were not significantly different between groups C and S after treatments. In manual ADT, there was no significant difference in the proportion of stable (grade I) to unstable (grade II and III) ankles between groups before treatment ($P = 0.74$), however, after treatment, the proportion of stable ankles was significantly higher in group S than in group C ($P = 0.02$) (Table 5).

4. Discussion

Although there has been extensive research into lateral ligament injuries associated with ankle sprains, the majority of research has focused on the ATFL, resulting in a lack of understanding of the CFL, particularly in relation to complete ruptures at the calcaneal attachment. This particular type of injury has been significantly understudied despite its clinical relevance. In this retrospective comparative study, we examine the outcomes of non-operative versus operative treatment of calcaneal CFL ruptures, with the aim of

Table 4
Preoperative Findings on MRI in Group C and S.

	Group C	Group S	P value
ATFL rupture sites			0.775
Fibular attachment	2 (14.3%)	5 (22.7%)	
Talar attachment	2 (14.3%)	4 (18.2%)	
Midsubstance	9 (64.3%)	10 (45.5%)	
No rupture	1 (7.1%)	3 (13.6%)	
CFL rupture sites			1
Calcaneal attachment	14 (100%)	22 (100%)	
PMD (mm)	11.2 $\pm$ 4.6	11.2 $\pm$ 5.3	0.958
Stener's-like lesion			0.126
No	10 (71.4%)	10 (45.5%)	
Yes	4 (28.6%)	12 (54.5%)	

ATFL, anterior talofibular ligament; CFL, calcaneofibular ligament; PMD, proximal migration distance

Table 5
Radiographic results in Group C and S.

	Group C		Group S		P-value
	Preoperative	Last follow-up	Preoperative	Last follow-up	
TTA (°)		6.5 ± 4.0		4.7 ± 0.4	0.491
ATD (mm)		5.4 ± 0.4		6.8 ± 0.5	0.599
Manual ADT (cases)					
Grade 0	0	2 (14.3 %)	0	14 (63.6 %)	
Grade I	1 (7.1 %)	3 (21.4 %)	1 (4.5 %)	6 (27.3 %)	
Grade II	7 (50 %)	5 (35.7 %)	10 (45.5 %)	2 (9.1 %)	
Grade III	6 (42.9 %)	4 (28.6 %)	11 (50 %)	0	
P-value			0.74 [†]	0.002 [‡]	

TTA, talar tilt angle; ATD, anterior talar displacement; ADT, anterior drawer test

[†]Difference in preoperative manual ADT between group C and S

[‡]Difference in manual ADT at last follow-up between group C and S

establishing a basis for a future randomised controlled trial to determine the optimal treatment approach.

The anatomical characteristics of ATFL and CFL differ; the ATFL is an expansion of a joint capsule with a broad membrane, while the CFL has a narrow, cord-like shape [2,3]. Therefore, unlike the ATFL, the CFL is more likely to be torn at its calcaneus or fibula attachment than at the midsubstance [17]. Kim et al. have elucidated that such injuries, particularly those marked by proximal migration, are not as rare as was previously conjectured. Within a cohort of 110 athletes, 52.7 % experienced complete CFL tears, with 62.1 % of these occurrences transpiring at the calcaneal insertion and 37.9 % presenting as sleeve-type avulsions. These findings suggest that calcaneal-side CFL injuries, although complex, require careful management to prevent long-term instability [18]. Lo et al. reported that displaced complete rupture of the ACL is associated with suboptimal anatomical healing and portends chronic knee instability [27]. It was also observed that, since the recovery from ligament-bone rupture was slower than that from ligament-ligament rupture, it required a more extended rehabilitation period [27]. Therefore, the complete rupture of calcaneal-side CFL takes longer to recover than a midsubstance rupture as it is a ligament-bone rupture.

Ultrasound has emerged as an increasingly favored diagnostic modality for the evaluation of soft tissue injuries, particularly those affecting the ankle ligaments [28]. Nevertheless, in instances of severe ankle sprains characterized by a complete rupture of the ATFL and CFL, it is frequently observed that the presence of edema and hematoma in the surrounding soft tissues complicates the accuracy of the diagnostic process. Furthermore, we have encountered discrepancies between ultrasound findings and MRI results in cases of complete rupture of the ATFL and CFL. Consequently, in clinical practice, if a CFL empty sign along with substantial hematoma or fluid accumulation within the peroneal tendon sheath is observed on ultrasound, indicating a severe injury, it is imperative to conduct an MRI to thoroughly evaluate the precise extent of ligamentous injury.

Unlike the ATFL, the CFL is an extraarticular structure not connected to a joint capsule. Furthermore, the fibular attachment of the CFL is connected to the ATFL by arciform fibers, while the calcaneal attachment has no connective structures [19]. Therefore, complete calcaneal-side CFL rupture can result in significant proximal migration of the ruptured stump. The presence of a gap resulting from proximal migration makes achieving anatomical healing challenging, ultimately leading to chronic instability. In particular, when a Stener's-like lesion occurs, where the ruptured stump is displaced outside of the peroneal tendons, anatomical healing is even more elusive and often requires surgical intervention [23]. Our results showed that although there was no significant difference regarding PMD and occurrence of Stener's-like lesion between groups C and S, the CAIT score was significantly better in group S at the last follow-up. Notably, in group C, the CAIT score fell below 24, indicating a persistently unstable ankle and an increased risk of recurrent injury.

Therefore, we recommend surgical repair in cases of complete calcaneal-side CFL rupture characterized by proximal migration from an anatomical origin.

Anatomically, the ATFL contributes primarily to ankle joint stability, while the CFL plays a critical role in ankle and subtalar joint stability [10,29]. While studies suggest that repair of the ATFL alone can yield good results [4,22], it's important to note that these findings generally apply to cases of chronic ankle instability rather than acute injuries. Our study, focusing on acute cases, highlights the need to address both ATFL and CFL tears, especially when there is significant proximal migration of the CFL. By repairing both ligaments in the acute phase, our approach aims to prevent the development of chronic instability and ensure long-term ankle stability, providing a more comprehensive solution. Thus, treatment approaches for isolated ATFL and combined ATFL and CFL injuries should differ. Although conservative treatment with functional rehabilitation has been advocated for most ankle sprains over the past decade [30,31], the optimal treatment of grade III injuries involving simultaneous ATFL and CFL tears remains controversial [11–13]. Surgical repair has been considered a viable option for patients aiming to return to high-demanding sports due to its potential to provide more excellent ankle stability and facilitate earlier RTS [12]. However, there has been limited understanding of the treatment and prognosis of CFL injuries.

When both ATFL and CFL ruptures occur, biomechanical instability becomes more pronounced [1,10,32]. When the CFL ruptures, it loses its role as a tensioner for the peroneal tendons, leading to decreased muscle function and potential functional abnormalities in the ankle. This loss of tension can compromise postural control and ankle stability, increasing the likelihood of developing chronic ankle instability. Samoto et al. comparatively analyzed the conservative treatment of isolated ATFL injuries and combined ATFL and CFL injuries [33]. They reported that the combined injuries had lower functional scores than the isolated ATFL injuries [33]. In addition, previous studies have suggested that combined injuries involving ATFL and CFL ruptures may also involve the interosseous talocalcaneal ligament rupture, leading to subtalar joint instability [10]. Consequently, it has been recommended that patients with combined ATFL and CFL injuries should undergo prolonged immobilization or consider surgical repair. When the ATFL and CFL rupture, it takes approximately 3 months to achieve RTS following appropriate immobilization and rehabilitation [34]. However, some patients with ankle injuries progress to mechanical instability with functional deficits during follow-up. In the present study, three (21.4 %) patients who were initially treated conservatively ultimately required surgical intervention owing to recurrent ankle sprains. This suggests that a complete calcaneal-side CFL rupture may result in an increased gap from the calcaneus, leading to increased laxity after improper healing, which is considered a cause of CAI.

However, the time required for ligament healing after ankle sprain remains uncertain. Systematic reviews have shown that significant improvements in mechanical stability do not occur until at least 6 weeks to 3 months after injury, with ankle laxity improving over 6 weeks to 1 year [34]. In addition, 7–42% of the patients still had objective mechanical laxity and subjective ankle instability up to 1 year after injury [34]. Another meta-analysis study evaluated the time to RTS after surgery in patients diagnosed with CAI and reported an average of 16.5 weeks [35]. In the present study, the time to RTS after ankle sprain with complete calcaneal-side CFL rupture was 17–18 weeks, and there was no significant difference between conservative and surgical treatment. However, conservatively treated patients were more dissatisfied with their treatment than surgically treated patients due to residual instability and recurrent sprains. In addition, 21% of the patients initially treated conservatively eventually underwent surgery for recurrent sprains. Although surgical treatment did not accelerate the onset of RTS, it was highly effective in preventing recurrent sprains. Therefore, surgical treatment of ankle sprains with complete calcaneal-side CFL rupture is considered a valuable approach, particularly in elite athletes, for whom recurrence after injury can be detrimental.

The strength of this study is that it is the first to compare the outcomes of conservative and surgical treatments specifically targeting complete calcaneal-side CFL rupture. Given the limited information available on treatment methods and prognosis of calcaneal CFL ruptures, this study is expected to provide valuable insights. In addition, this study accurately evaluated the proximal migration of the CFL and the occurrence of Stener's-like lesion by performing an MRI in the 45° oblique coronal plane for calcaneal-side CFL rupture [26]. However, this study has several limitations. First, it included a relatively small number of participants. Although the incidence of calcaneal-side CFL rupture is not well known, including 36 cases is considered substantial, especially considering the relative rarity of this condition [22]. Second, the study was conducted retrospectively, which may introduce bias. Although no significant difference in the proximal migration of the distal end of the ruptured CFL was observed between the two groups, surgeons may have been more inclined to operate in cases with severe proximal migration or occurrence of Stener's-like lesion. Third, our conservative group may not be fully comparable to the operative group because of potential selection bias based on clinical judgement. Imaging helped to assess injury severity, but factors such as patient activity, pain tolerance and previous injuries may have influenced treatment decisions. It's important to note that not all severe CFL injuries require surgery; many patients, including athletes, recover well with conservative treatment. Surgery should be reserved for cases with persistent instability or specific injury patterns. Therefore, a randomized prospective comparative study with a larger cohort of patients is warranted to minimize potential bias and provide further clarity regarding the optimal treatment approach.

5. Conclusions

Surgical treatment showed better clinical outcomes and increased stability than conservative management in ankle sprains with complete calcaneal-side CFL rupture. Therefore, surgical intervention is emerging as an effective strategy to prevent recurrence after injury and represents a favorable treatment choice, especially for highly active individuals in whom the prevention of recurrence is paramount.

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Declaration of Competing Interest

Authors declare that they have no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fas.2025.05.014.

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